Segurança de Sistemas e dados

Aula 3

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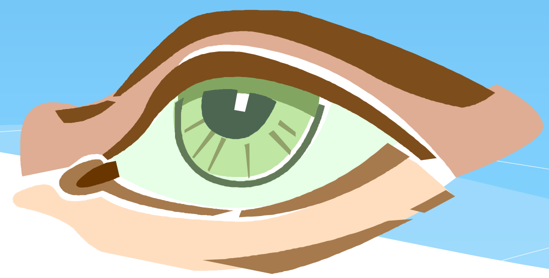
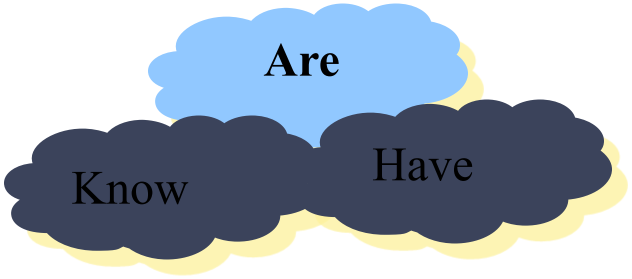
Biometrics



Something You Are

* Biometric
  * **"You are your key"** — Schneier
* Examples
  * Fingerprint
  * Handwritten signature
  * Facial recognition
  * Speech recognition
  * Gait (walking) recognition
  * "Digital doggie" (odor recognition)
  * Many more!



Something You Are

Worldcoin iris biometrics cryptocurrency project launched by Sam Altman

Oct 25, 2021, 3:03 pm EDT | Alessandro Mascellino

CATEGORIES Biometrics News | Financial Services | Iris / Eye Recognition

Tech millionaire Sam Altman's Worldcoin start-up has emerged from stealth mode, offering free cryptocurrency to individuals who verify their accounts by taking an iris scan via a dedicated biometric device.

At the time of writing, and days after its official launch, Worldcoin's valuation has already reached $1 billion, after substantial investments from Andreessen Horowitz, Coinbase, and LinkedIn Co-founder Reid Hoffman.

The start-up was founded by Altman, together with theoretical physics student Alex Blania and investment expert Max Novendstern.

Worldcoin has already shipped its orb-shaped iris biometric devices to testers in 12 countries, with over 100,000 users globally already, and reportedly 700 more each week.

Biometric Authentication

* authenticate user based on one of their physical characteristics

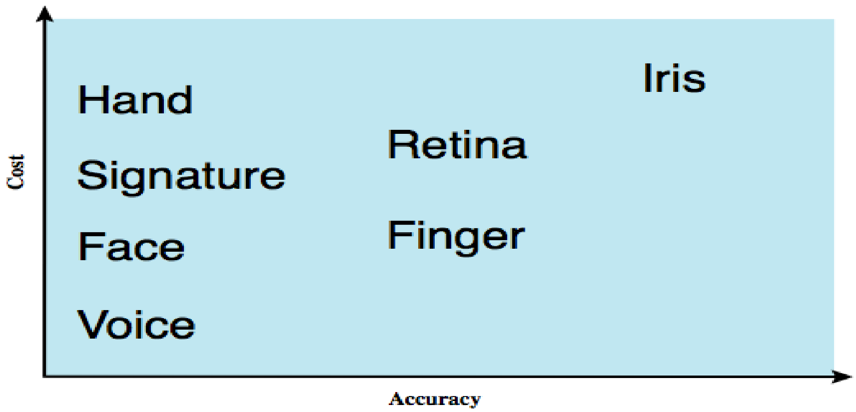


Figure 3.6 Cost versus accuracy of various biometric characteristics in user authentication schemes.

Ideal Biometric

* **Universal** — applies to (almost) everyone
  * In reality, no biometric applies to everyone
* **Distinguishing** — distinguish with certainty
  * In reality, cannot hope for 100% certainty
* **Permanent** — physical characteristic being measured never changes
  * In reality, OK if it to remains valid for long time
* **Collectable** — easy to collect required data
  * Depends on whether subjects are cooperative
* Safe, user-friendly, etc., etc.

Biometric Modes

* **Identification** — Who goes there?
  * Compare **one-to-many**
  * Example: The FBI fingerprint database
* **Authentication** — Are you who you say you are?
  * Compare **one-to-one**
  * Example: Thumbprint mouse
* Identification problem is more difficult
  * More "random" matches since more comparisons
* We are interested in authentication

Cooperative Subjects?

- Authentication — cooperative subjects
- Identification — uncooperative subjects
- For example, facial recognition
  - Used in Las Vegas casinos to detect known cheaters (terrorists in airports, etc.)
  - Often do not have ideal enrollment conditions
  - Subject will try to confuse recognition phase
- Cooperative subject makes it much easier
  - We are focused on authentication
  - So, subjects are generally cooperative

Biometric Accuracy

- never get identical templates
- problems of false match / false non-match

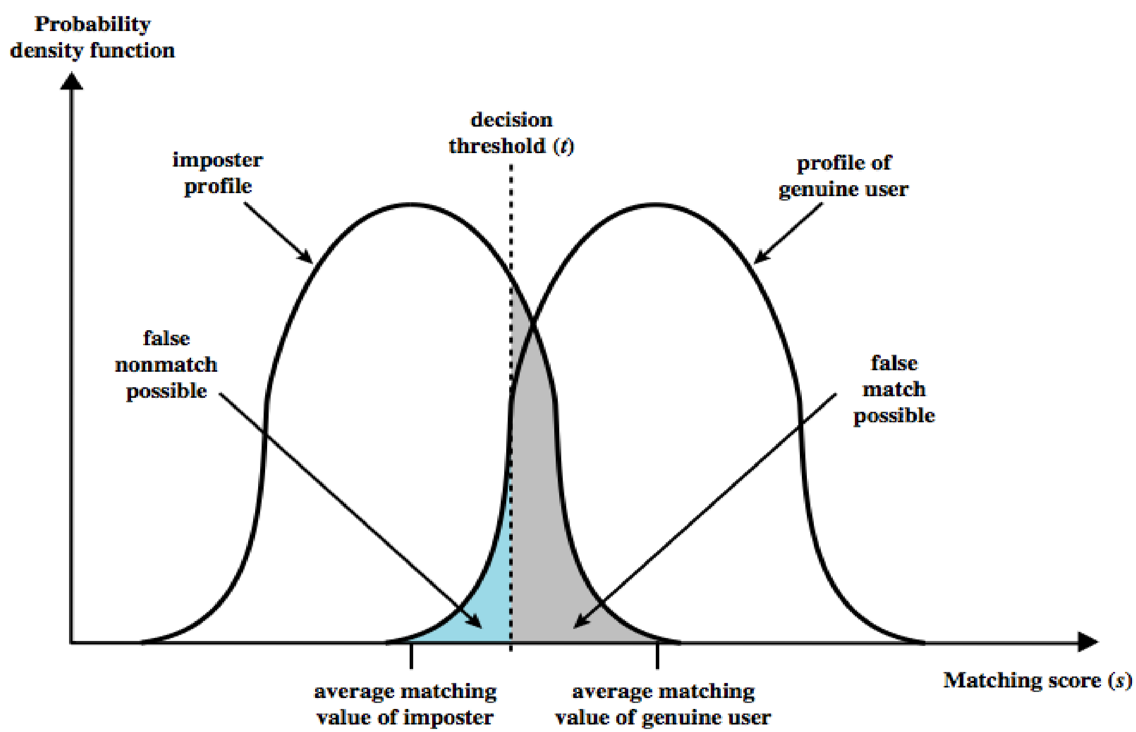


Figure 3.8 Profiles of a biometric characteristic of an imposter and an authorized users. In this depiction, the comparison between presented feature and a reference feature is reduced to a single numeric value. If the input value (s) is greater than a preassigned threshold (t), a match is declared.

Biometric Errors

* Fraud rate versus insult rate
  * **Fraud:** Trudy mis-authenticated as Alice **(False Positive (FP))**
  * **Insult:** Alice not authenticated as Alice **(False Negative (FN))**
* For any biometric, can decrease fraud or insult, but other one will increase
* For example
  * 99% voiceprint match: low fraud, high insult
  * 30% voiceprint match: high fraud, low insult
* **Equal error rate: rate where fraud == insult**
  * A way to compare different biometrics

Biometric Accuracy

* can plot characteristic curve
* pick threshold balancing error rates

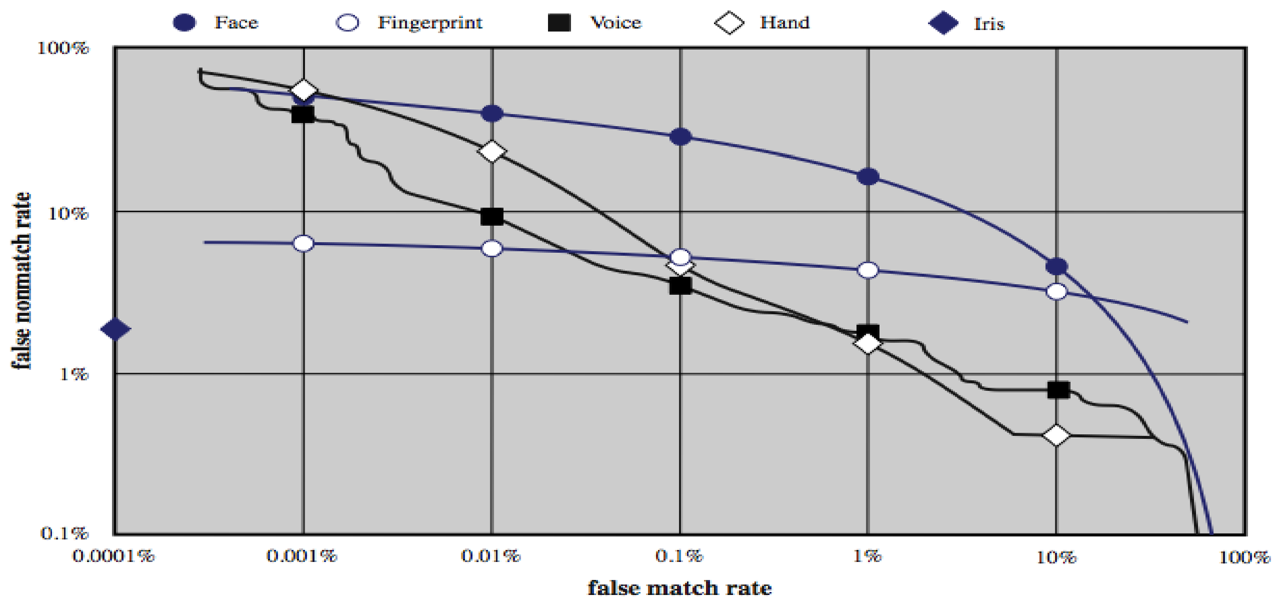


Figure 3.10 Actual biometric measurement operating characteristic curves, reported in [MANS01]. To clarify differences among systems, a log-log scale is used.

Enrollment vs Recognition

- Enrollment phase
  - Subject's biometric info put into database
  - Must carefully measure the required info
  - OK if slow and repeated measurement needed
  - Must be very precise
  - May be weak point of many biometric
- Recognition phase
  - Biometric detection, when used in practice
  - Must be quick and simple
  - But must be reasonably accurate

Operation of a Biometric System

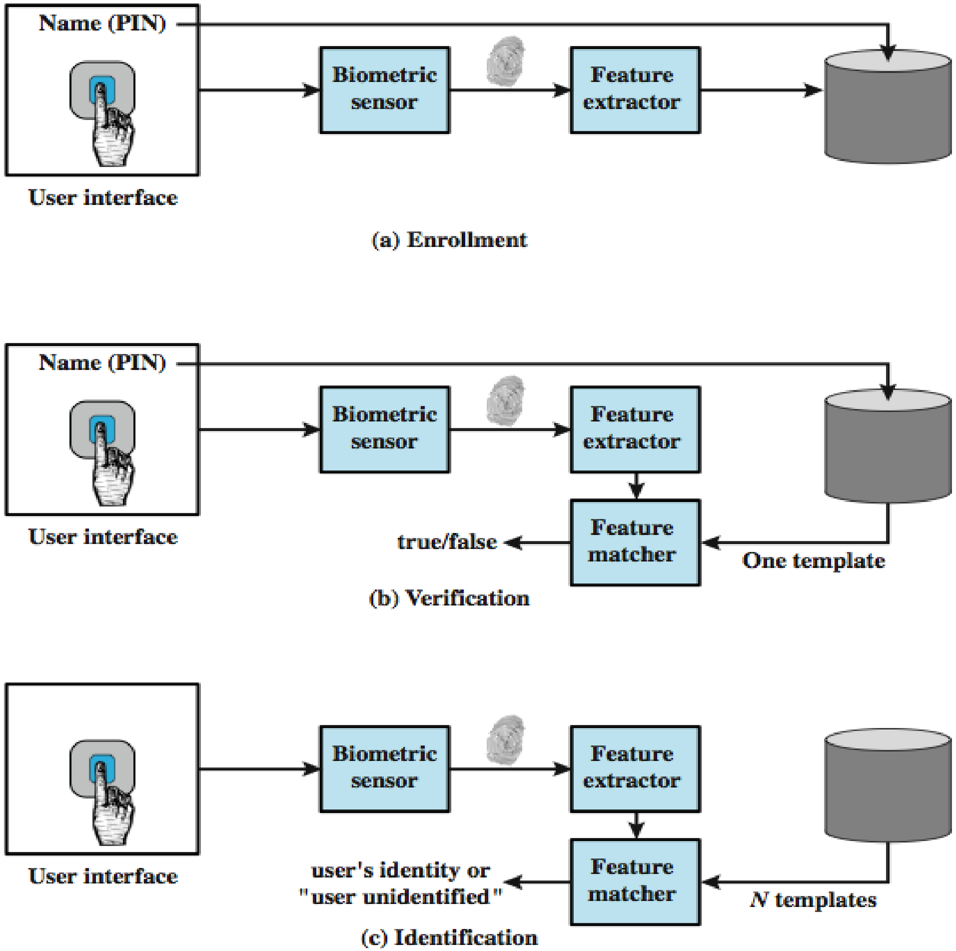


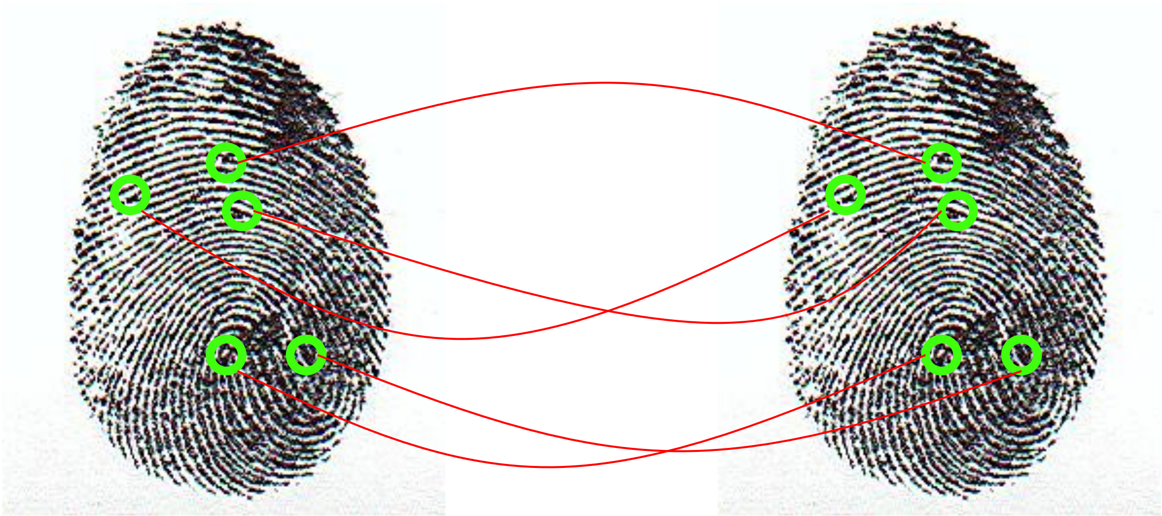
Figure 3.7 A generic biometric system. Enrollment creates an association between a user and the user's biometric characteristics. Depending on the application, user authentication either involves verifying that a claimed user is the actual user or identifying an unkown user.

Fingerprint: Enrollment

- Capture image of fingerprint
- Enhance image
- Identify points

Fingerprint: Recognition

* Extracted points are compared with information stored in a database using statistical matching.



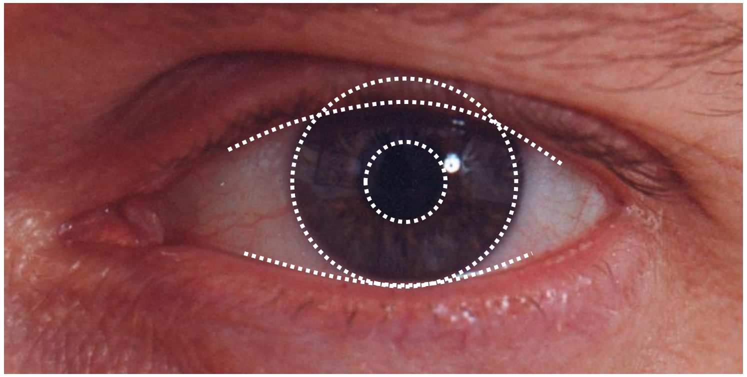
Hand Geometry

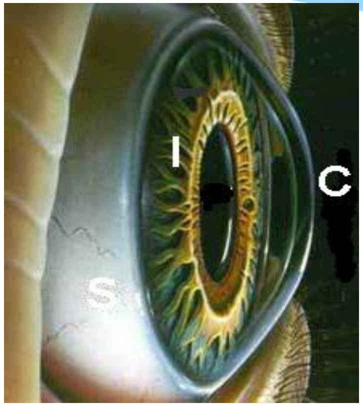
- A popular biometric
- Measures shape of hand
  - Width of hand, fingers
  - Length of fingers, etc.
- Human hands not unique
- Hand geometry sufficient for many situations
- OK for authentication
- Not useful for ID problem

Hand Geometry

- Advantages 
  - Quick — 1 minute for enrollment, 5 seconds for recognition
  - Hands are symmetric — so what?
- Disadvantages 
  - Cannot use on very young or very old
  - Relatively high equal error rate

Iris Patterns





* Iris pattern development is "chaotic"
* Little or no genetic influence
* Different even for identical twins
* Pattern is stable through lifetime

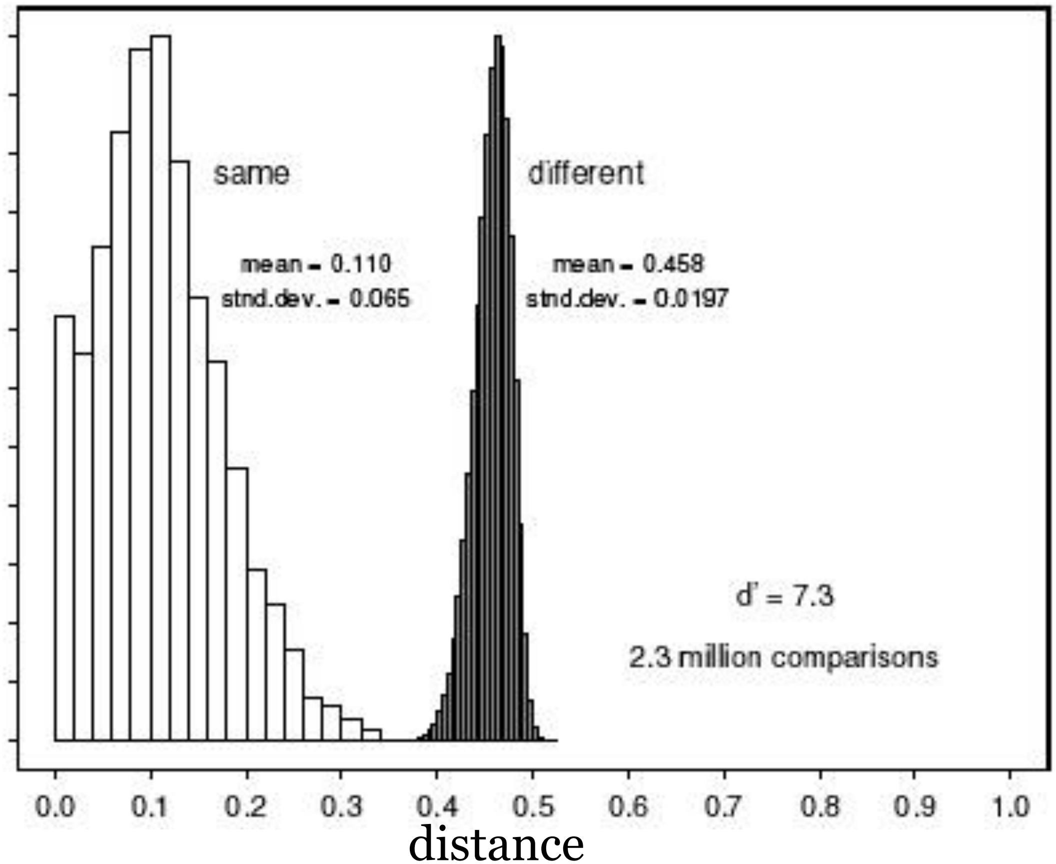
Iris Scan Error Rate

$$distance = \frac{\#bits\ that\ don't\ match}{\#bits\ compared}$$

distance	Fraud rate
0.29	1 in $1.3*10^{10}$
0.30	1 in $1.5*10^{9}$
0.31	1 in $1.8*10^{8}$
★ 0.32	1 in $2.6*10^{7}$
0.33	1 in $4.0*10^{6}$
✶ 0.34	1 in $6.9*10^{5}$
0.35	1 in $1.3*10^{5}$

✶ equal error rate

★ usually used



Equal Error Rate Comparison

- **Equal error rate (EER): fraud == insult rate**
- **Fingerprint** biometric has **EER of about 5%**
- **Hand geometry** has **EER of about 10-3**
- **In theory, iris scan has EER of about 10-6**
  - But in practice, may be hard to achieve
  - Enrollment phase must be extremely accurate
- **Most biometrics much worse than fingerprint!**
- **Biometrics useful for authentication…**
  - …**but identification biometrics almost useless today**

Biometrics: The Bottom Line

* Biometrics are hard to forge
* But attacker could
  * Steal Alice’s thumb
  * Photocopy Bob’s fingerprint, eye, etc.
  * Subvert software, database, “trusted path” …
* And how to revoke a “broken” biometric?
* **Biometrics are not foolproof**
* Biometric use is limited today
* That should change in the (near?) future

Some devices

Something you have

Tokens

Token Authentication

- object user possesses to authenticate, e.g.
  - embossed card
  - Bank random matrix codes plastic cards
  - magnetic stripe card
  - memory card
  - Smartcard
  - Smartphone App

Should be hard to copy and/or well protected.

Memory Card

* store but do not process data
* magnetic stripe card, e.g. old bank cards
* electronic memory card
  * Old phone cards
* used alone for physical access
* with password/PIN for computer use
* drawbacks of memory cards include:
  * need special reader
  * loss of token issues
  * user dissatisfaction

Smartcard

- credit-card like
- has own processor, memory, I/O ports
  - wired or wireless access by reader
  - may have crypto co-processor
  - ROM, EEPROM, RAM memory
- executes protocol to authentic
  reader/computer
- also have USB dongles

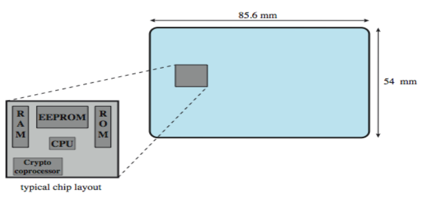


Figure 3.4 Smartcard dimensions. The smartcard chip is embedded into the plastic card and is not visible. The dimensions conform to ISO standard 7816-2

Cartão cidadão

* BI
* Segurança Social
* Serviço Nacional de Saúde
* Número de contribuinte
* Cartão de Eleitor
* Documento de viagem.

2-factor Authentication

- Requires any 2 out of 3 of
  - Something you know
  - Something you have
  - Something you are
- Examples
  - ATM: Card and PIN
  - Credit card: Card and signature
  - Password generator: Device and PIN
  - Smartcard with password/PIN

Problems with Passwords Revisited

- Selection
  - Good secure passwords are hard to find
- Memorization
  - It is easy to forget infrequently used passwords
  - It is hard to remember secure passwords
- Reuse
  - To many different passwords to memorize
  - People reuse the same password all over

Problems with Passwords in clinical settings

* In hospitals it is common to find post-it notes with application passwords.
* Health professionals often share accounts with each other.
    * To cover each other backs.
    * To circumvent hardly usable security
    * To delegate tasks to interns.
    * To save lives. (badly designed RBAC models and hard to use security mechanisms)
* Keyboards are also among the dirtiest objects in the planet

Problems with Passwords Revisited

- Sharing
  - When working in groups it is common for people to share passwords
  - Non repudiation is easy to circumvent with password sharing
  - When people do not assess the real value of their digital assets they share passwords.
- Malware (Virus, Trojans)
  - Password Sniffing, Phishing Attacks, etc...

Problems with Passwords Revisited

- Online Banking is one prime example of management of highly valuable assets on the Internet.
  - Online Banks are very convenient for the costumer and save of lot money to the Banks.
  - "Phishing" attacks became widespread and are quite effective at stealing user credentials.
  - Banking dedicated Malware provides high returns to the attacker
- Risk Analysis tell us that Login/Password is not appropriate to protect these assets.

Classical Factor Two Authentication tokens

- The classic solution has been to employ factor two (difficult to clone) authentication

Classical Factor Two Authentication tokens

* Users need to carry with them additional physical tokens and/or readers.
* Generally, requires specific drivers and middleware software installed on users' workstations.
* Are proprietary in Nature and incompatible with each other.
  * Single purpose (Generally you cannot use Institution A token to authenticate yourself into Institution B).

More Flexible Factor Two Authentication tokens

* Yubikeys:
  * Low cost one time password (OTP) generator token (40 chars).
  * Connects to USB port.
  * Acts like a keyboard, no driver required.
  * Press button with your finger to generate a new OTP.
  * Very easy to integrate with legacy login/password authentication schemes.
  * Token validation service on the cloud.
  * Widely deployed by well know Internet companies (Paypal, Google, LastPass,…)
  * Direct support currently being integrated into Google chrome for a more seamless authentication with HTML5 sites.

More Flexible Factor Two Authentication tokens

- Yubikeys:
  - Current Version has two slots. Each can store 128bits.
  - These can be used in several modes:
    - Yubikey OTP
    - OAUTH-HOTP (RFC 4226)
    - Static 128 bit password
    - Challenge Response
  - Newer models are NFC enabled.
    - Yubikey NEO
  - Yubikey software is mostly open Source.
  - For the server side Yubico also supply a low cost HSM to securely protect shared secrets.

Yubico OTP

https://developers.yubico.com/OTP/OTPs_Explained.html